Network Security and Cryptography – August 29th, 2022

Warning

The solutions in this exam paper have not been reviewed by the lecturer and contain errors. If you find any errors, please let me know.

Exercise 1

Elliptic Curves Addition

Let $y^2 = x^3 + bx + c \bmod p$ and $P_3 = P_1 + P_2$

$$m = \begin{cases} (y_2 - y_1)(x_2 - x_1)^{-1} \bmod p & \text{if } P_1 \neq P_2 \\ (3x_1^2 + b)(2y_1)^{-1} \bmod p & \text{if } P_1 = P_2 \end{cases}$$

$$x_3 = m^2 - x_1 - x_2 \bmod p$$

$$y_3 = m(x_1 - x_3) - y_1 \bmod p$$

Consider the Diffie-Hellman key exchange over the Elliptic Curve:

$$E : y^2 = x^3 + x + 1 \bmod 23$$

with $N = 28$ points. Point $P = (1, 7)$ is a generator.

Alice chooses a secret number a and sends the point $A = aP = (18, 20)$. Bob chooses a secret number $b = 6$ and sends the point $B = bP = (12, 19)$. At the end of the protocol, Alice and Bob calculate the secret point S.

1. Verify that the curve is not singular.

Solution:

$$4 \times 1^3 + 27 \times 1 = 31 \neq 0 \pmod{23}$$

not singular.

2. Write the points for $x = 0$ and $x = 2$.

Solution:

Consider $x = 0$. We need to solve $y^2 = 1 \bmod 23$. Since

$$\left(\frac{1}{23}\right) = 1$$

there are two roots: $y = \pm 1$. The two points are $(0, 1), (0, 22)$.

Consider $x = 1$. We need to solve $y^2 = 3 \bmod 23$. Since

$$\left(\frac{3}{23}\right) = 3^{11} \bmod 23 = 1$$

there are two roots.

Now,

$$3^6 \bmod 23 = 16$$

The two points are $(1, 16), (1, 7)$.

3. Verify that the order of A is not 4.

Solution:

We need to verify that $4A \neq \infty$.

We first compute $2A = (12, 19)$. Then $4A = 2(2A) = (5, 19) \neq \infty$.

4. Compute the expected number of point operations necessary to compute $S = bA$ using Double & Add.

Solution:

The number of operations is $\frac{3}{2}\lceil \log_2 b \rceil \simeq 1.5 \log_2 N$.

5. Compute the number of point operations that an attacker expects to spend to find b using exhaustive search.

Solution:

We add P to itself multiple times until we find B. On average the attacker needs $N/2$ attempts.

In a new execution of the protocol, Alice publishes a new point A having order 4.

6. What can we say about the value of a?

Solution:

We known that $4A = 4aP = \infty$. We also know that $NP = \infty$, then a possible solution is $4a = N$, $a = N/4$.

7. How many unique values of S can be obtained at the end of the protocol execution?

Solution:

The number of possible values of S is equal to $\text{ord}(A)$.

8. Write the `python` commands to generate the public and the private key on the NIST curve SECP384r1 curve.

Solution:

```
private_key = ec.generate_private_key( ec.SECP384R1() )
```

Exercise 2

An attacker obtains a user database containing a list of $N = 10^7$ usernames along with the corresponding salts and stored hashes. Each user password is concatenated to an 80-bit salt and hashed with hash function h, which outputs 100 bits. One third of the users picked a password that is included in a wordlist of $L = 100{,}000$ common passwords. An attacker can hash $w = 10{,}000$ passwords per second and works for $T = 1$ week.

9. What is the probability that any two random users have the same common password?

Solution:

We assume that the users with uncommon passwords all have unique passwords. The users with common passwords are $N/10 = 10^6$, who pick their passwords in a set of L, which is smaller than N. By virtue of the pidgeonhole principle, the probability that there are two users with the same password is 1.

10. What is the probability that any two random users have the same hash?

Solution:

The passwords are salted, which means tha all inputs to the hash function are unique. Consequently, the probability is:

$$1 - \exp\left(-\frac{N^2}{2^{101}}\right) = 3.9 \times 10^{-17}$$

11. Calculate the probability that the attacker can recover the password of a random user.

Solution: The attacker can easily test all the passwords in the wordlist for a given user. Therefore the attack is successful if the user's password is a common one:

$$\Pr\{\text{finding password}\} = \frac{1}{10}$$

12. Calculate the expected number of passwords that the attacker can find in time T.

Solution:

Testing all the common passwords for a single user takes a time $\tau = L/w$ In a time T the attacker can test T/τ users.

Assuming that $T/\tau < N$, the average number of passwords found is:

$$\frac{1}{10}\frac{T}{\tau} = \frac{Tw}{10L} = 60{,}480$$

13. Calculate the amount of password stretching that makes the attacher find, on average, 0.1 passwords in time T.

Solution: The stretching decreases the hash rate by a factor r.

$$\frac{Tw}{10Lr} = 1$$

Then, $r = 60{,}480$.

14. Consider a salt size of 16 bits. What is the probability that there exist two users having the same salt?

Solution:

$$Because of the pigeonhole principle, the probability is 1.$$

Exercise 3

Alice and Bob play heads or tails over the phone. The full protocol is as follows:

a) Some trusted third party provides the elliptic curve E, which has $N = 66$ points, a base point G with order q and the point $H = aG$. The value a is secret.

b) Alice chooses a binary value x and a random number r. She outputs the point $C = xG + rH$.

c) Bob chooses the binary value y and publishes it.

d) Alice reveals x and r.

e) Alice wins if $x = y$, she loses otherwise.

In a specific execution of the protocol, Alice chooses $x = 0, r = 2$. Bob publishes his choice $y = 1$.

15. What are the possible values for q?

Solution: 2, 3, 6, 11, 22, 33, 66

16. Consider the case in which q is the largest prime factor of N. How many unique values can the point C take?

Solution: q

17. Calculate the probability that a random value r' can be used by Alice to reveal $x = 1$ and cheat.

Solution: The probability is $1/q$

18. Suppose that the value $a = 4$ is leaked. Calculate a value r' that Alice can use to cheat.

Solution: We need to solve the equation:

$$2H = C = G + r'H$$

In general, this is hard. However, we know that $H = 4G$, so we can write:

$$8G = G + 4r'G$$

which becomes:

$$7 = 4r' \bmod 11$$

from which $r' = ...$